

Analysis Report

1 Treatment Effect on Hiring Outcomes

1.1 Blinding and Demographic-Based Premiums / Penalties

- **Blinding disrupts pre-existing system of demographic premiums and penalties.**
 - **Winners: Hispanic Female in Resume-Screening (Phase 1) and White Male in Technical-Evaluation (Phase 2).** Blinding removes the baseline *penalty* these groups received under non-blind evaluations. Their evaluation scores rise by **+8.7%** and **+8.6%** of a SD respectively.
 - **Losers: Hispanic Male in Phase 1 and Hispanic Female in Phase 2.** Blinding removes the baseline *premium* these groups received under non-blind evaluations. Their evaluation scores fall by **-11.5%** and **-6.9%** of a SD respectively.
- **Caveat:** Hiring is often an *ordinal ranking* problem — one candidate’s gain corresponds to another’s lost opportunity — so the absence of statistical significance on a given row does not mean the corresponding group is unaffected by blinding.

Table 1: Effect of Blinding on Overall Evaluation Score by Race-Gender

	Panel A: Resume-Screening	Panel B: Technical-Evaluation
	HR	Engineer
White Male	0.037 (0.058)	0.086*** (0.028)
White Female	0.013 (0.051)	0.024 (0.036)
Asian Male	0.014 (0.059)	-0.017 (0.029)
Asian Female	-0.023 (0.048)	-0.030 (0.029)
Hispanic Male	-0.115** (0.048)	-0.004 (0.030)
Hispanic Female	0.087* (0.052)	-0.068** (0.034)
Display Position FE	✓	✓
Resume ID FE	✓	✓
Resume - Coding Test Version FE	×	×
Respondent ID FE	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓
Observations	2466	2754
R ²	0.423	0.749
Adj. R ²	0.409	0.744

(1) * p < 0.1, ** p < 0.05, *** p < 0.01

(2) Standard errors clustered by respondent.

1.1.1 Heterogeneity: Rejecting Traditional In-Group Favoritism

- **Findings:** Our sample of software engineers demonstrates a **preference for the *out-group*** (i.e., a preference for *under-represented groups*).
 - Under blinding, the scores of *concordant* (race/gender) applicants evaluated by Engineers rise by **+6.9%** of a SD ($p < 0.05$), while the scores of *discordant* applicants fall by **-0.9%** of a SD ($p < 0.05$).
 - **White Males** make up the majority of the evaluator pool for engineers, and their behavior *drives the aggregate results*.
- **Takeaway:** The **majority gatekeepers** were *actively penalizing* candidates who looked like themselves and assigning a *premium to evaluation scores of minority candidates* when such demographic information is visible. Enforcing blind hiring stops the majority group, **White Males**, from systematically boosting the minority candidates. This explains why, in our experiment, **White Male applicants are the primary beneficiaries** of blind hiring: they are “rescued” from the baseline *penalty* imposed by their own demographic engineer peers.

Table 2: Effect of Blinding on Overall Evaluation Score by Evaluator-Candidate Concordance (Race / Gender / Race+Gender)

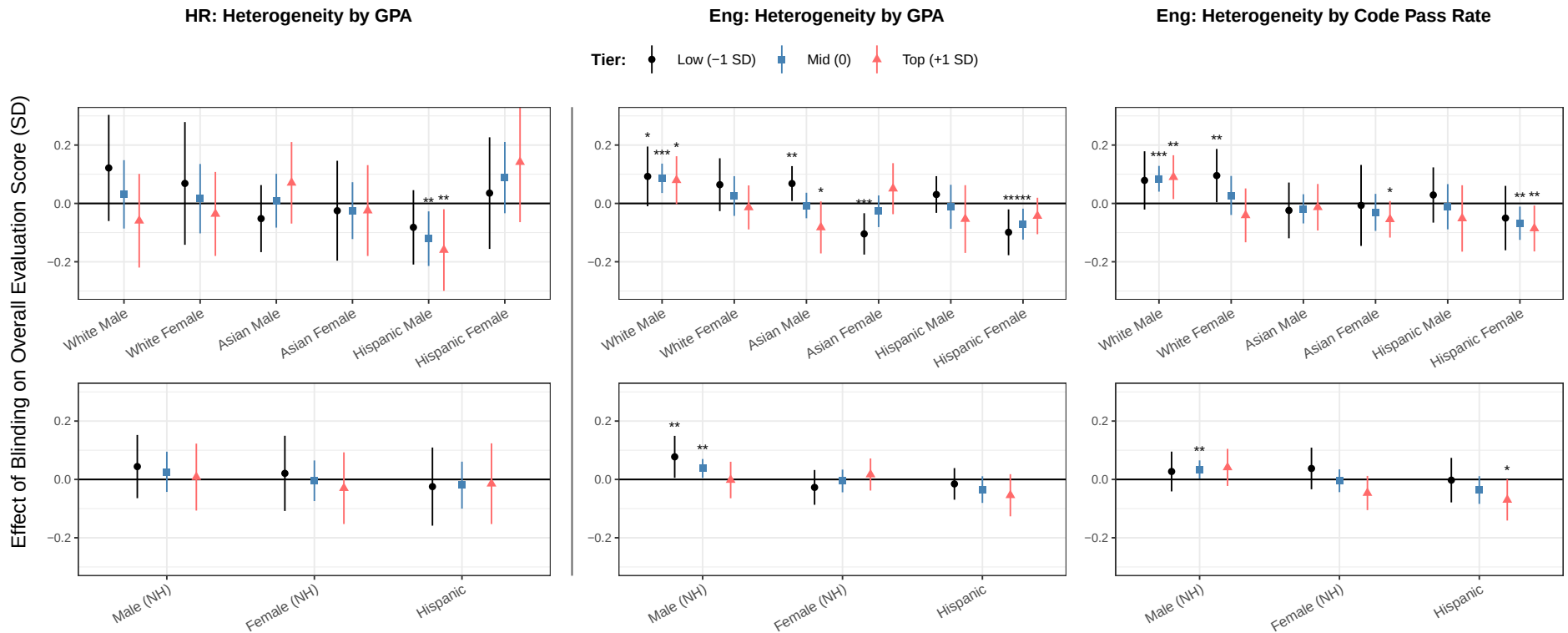
	Panel A: Resume-Screening			Panel B: Technical-Evaluation		
	Race	Gender	Race+Gender	Race	Gender	Race+Gender
Concordant Applicant	-0.020 (0.042)	-0.010 (0.027)	-0.076 (0.068)	0.045* (0.023)	0.029* (0.015)	0.069** (0.032)
Nonconcordant Applicant	0.007 (0.015)	0.009 (0.025)	0.011 (0.010)	-0.014* (0.007)	-0.029* (0.016)	-0.010** (0.004)
Display Position FE	✓	✓	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×	×	×
Respondent ID FE	×	×	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓	✓	✓
Observations	2466	2466	2466	2754	2754	2754
R ²	0.421	0.421	0.421	0.748	0.749	0.748
Adj. R ²	0.408	0.408	0.408	0.743	0.743	0.743

(1) * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

(2) Standard errors clustered by respondent.

1.1.2 Heterogeneity: Where the Demographic Premium Applies (Effects by Candidate Quality)

- **Findings:** The blinding effect on evaluation scores diverges across visible candidate quality — GPA (both phases) and the code pass rate (Engineer phase only, as HR never saw the code) — within the same demographic group. The implicit out-group premium documented in §1.1.1 is concentrated at the *top* of the ability distribution.
- **Minorities.** In the *resume-screening* phase, blinding removes the **premium Hispanic Males** previously received at mid and high ability tiers, as defined by the visible ability cue (**GPA**). In the *technical-evaluation* phase, blinding hurts the **gender and racial minorities**, in particular **Hispanic Females**, by removing the **premium** they previously received under nonblind.
- **Majority (White Male mirror).** Blinding raises **White Male** evaluation scores roughly *uniformly* across all ability tiers in the Engineers' *technical-evaluation* phase.
- **Takeaway.** Evaluators' demographic adjustments under nonblind are *threshold-based*. **Minorities** receive a **premium only when** they clear a high visible-ability bar — concentrated at the *top* of the GPA / code-success-rate distribution. Evaluators appear to want to advance minority candidates **but not at the cost of quality**, reserving the premium for the strongest minorities and giving nothing to the rest. The mirror **penalty** applied to **White Males** seems to be *unconditional*, a flat shift across ability tiers. Blinding removes both adjustments simultaneously.

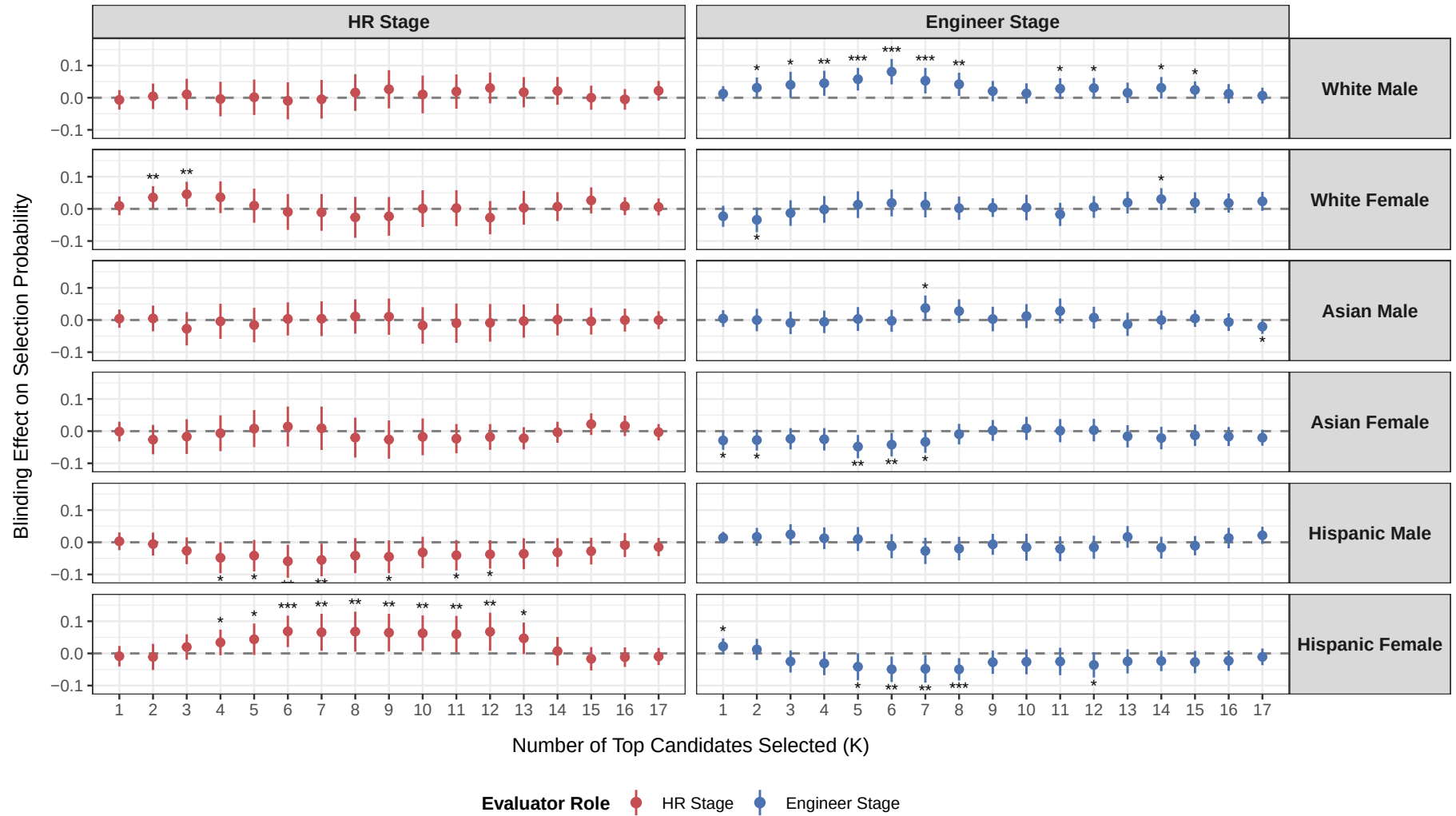


1.2 Blinding Reshuffles Demographics of Hired (Composition Effect)

- **Findings:** Mirroring the per-resume rating effects in §1.1, blinding **reshuffles who advances** at each phase.
 - *Resume-screening phase:* **Hispanic Females** gain a higher pass rate to the next round; **Hispanic Males** lose.
 - *Technical-evaluation phase:* **White Males** gain at the hiring stage; **Asian Females** and **Hispanic Females** lose.

Effect of Blind Hiring on Selection Probability by Demographic Group

Blinding effect = $-(\text{effect of revealing identity vs blind})$. Positive = more likely selected under blinding.



Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Base group: Blind resumes.

2 Allocation of Attention and Material Engagement

2.1 Determinants of Attention (Descriptive)

- **Determinants of Time Spent on Evaluation.** There is **no evidence** that candidate quality or candidate demographics drive how much time an evaluator spends evaluating a candidate. The only meaningful determinant is **display-order learning or fatigue**: even after removing the first three candidates evaluated as warm-up, each later position receives less time (HR $-0.5\text{s}/\text{position}$ on a $\sim 40\text{s}$ mean, $\sim 17\%$ cumulative; Engineers -1.4s on a $\sim 57\text{s}$ mean, $\sim 34\%$; both $p < 0.01$).

Table 3: Determinants of Time on Page (Seconds)

	Outcome: Time on Page (Seconds per Candidate)	
	Panel A: Resume-Screening	Panel B: Technical-Evaluation
	(1)	(2)
GPA (Z)	-0.83 (0.60)	-0.55 (1.21)
# Awards (Z)	0.01 (0.68)	0.57 (1.03)
Top Experience	-0.89 (1.41)	3.95^* (2.26)
Test Cases Passed (Z)		-0.16 (1.66)
Quality Mismatch I (Strong Resume, Weak Coding)		-0.84 (4.08)
Quality Mismatch II (Strong Coding, Weak Resume)		3.50 (2.66)
Display Position	-0.50^{***} (0.16)	-1.40^{***} (0.23)
Mean Time (Seconds)	40.1	57.2
Display Position FE	$\times$	$\times$
Resume ID FE	$\times$	$\times$
Resume - Coding Test Version FE	$\times$	$\times$
Respondent ID FE	$\checkmark$	$\checkmark$
Respondent Age (Cont) and Demographic FEs	$\times$	$\times$
Observations	2050	2290
R ²	0.286	0.464
Adj. R ²	0.233	0.424

(1) * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

(2) Standard errors clustered by respondent.

(3) The sample uses evaluations from display positions 4 – 18; the first three candidates evaluated are removed because early evaluations reflect warm-up (interface acclimation) patterns, not steady-state attention.

(4) Quality Mismatch I (Strong Resume, Weak Coding) and II (Strong Coding, Weak Resume) split candidates by whether the paper composite (GPA, awards, firm prestige) and the test-case pass rate fall on opposite sides of their means; reference = concordant candidates. Engineer-only.

(5) Assigned race/gender, entered jointly with the quality controls (nonblind, White Male omitted), does not predict time in either phase (joint $p = 0.54$ HR, 0.34 Engineer).

- **Determinants of Material Engagement.** Engagement, by contrast, *does* track candidate quality — but only for **opening the full code, not the full resume**. Engineers open a candidate’s code more when the candidate has a higher GPA (+1.5 pp per SD), a higher test-case pass rate (+1.7 pp), or a strong code pass rate paired with a relatively weak resume — the “**hidden gem**” (+3.2 pp, ~+5% on a ~64% mean; all $p < 0.05$). Resume-opening did not respond to any of these.

Table 4: Determinants of Opening Rates (Engineer Phase Only)

	Outcome: Opened the File (1 = opened)	
	Open Resume	Open Code
GPA (Z)	0.7 (0.7)	1.5** (0.7)
Test Cases Passed (Z)	0.8 (0.6)	1.8** (0.8)
# Awards (Z)	-0.4 (0.6)	0.8 (0.6)
Top Experience	0.5 (1.2)	0.4 (1.1)
Quality Mismatch I (Strong Resume, Weak Coding)	-0.1 (1.6)	1.9 (1.7)
Quality Mismatch II (Strong Coding, Weak Resume)	0.8 (1.2)	3.3** (1.5)
Mean Open Rate (%)	64.7	64.3
Display Position FE	×	×
Resume ID FE	×	×
Resume - Coding Test Version FE	×	×
Respondent ID FE	✓	✓
Respondent Age (Cont) and Demographic FEs	×	×
Observations	2,754	2,754
R ²	N/A	N/A
Adj. R ²	N/A	N/A

(1) * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

(2) Standard errors clustered by respondent.

(3) For this table, each coefficient cell is its own separate one-signal regression, with the one exception that the two Quality-Mismatch flags share a single regression (entered jointly). This split is deliberate: entered jointly, collinearity across the 18 candidates kills the individual coefficients.

2.2 Treatment Effects on Engagement (Causal)

- **Effect of Blinding on Time Spent.** Blinding does **not** change how long evaluators spend per candidate in either phase — no effect for HR or Engineers.
- **Effect of Blinding on Material Engagement.** It does change *what* engineers engage with: under blinding they **open a candidate’s code** $\sim +11$ pp more often (on a $\sim 64\%$ base), while resume-opening is unchanged. This boost is **uniform** — interacting it with candidate characteristics (test cases, GPA, awards, top-firm experience, the two quality-mismatch flags) shows no moderation by candidate quality (all $p > 0.22$).
- **Blinding changes how engineers work.** The extra opening *is* modestly informative — among the **strongest candidates**, where the free coding-ability signal (the test-case pass rate) saturates: there the engineer’s coding rating tracks true coding ability only weakly ($r \approx 0.31$), and opening the code lifts it to $r \approx 0.41$. Yet hiring efficiency is null (explored in Section 6), because blinding raises opening *uniformly* rather than aiming it at that margin, changes the opening rate only *modestly*, and the gain there is far too small to reorder the candidate rankings.

Table 5: Effect of Blinding on Attention & Engagement

	Time (sec)		Opened, Eng. only (pp)	
	HR	Eng.	Op. Res.	Op. Code
Nonblind Mean	40.3	56.8	61.7	58.4
Blinding Effect	-0.4 (3.1)	+2.2 (5.2)	+5.5 (6.0)	+11.6** (5.8)
Display Position FE	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓
Test-Version FE	×	×	×	×
Respondent ID FE	×	×	×	×
Resp. Age & Demog. FEs	✓	✓	✓	✓
Observations	2,050	2,290	2,754	2,754

Table 6: Heterogeneity of the Effect by Candidate Quality

Blinding ×	Time (sec)		Opened, Eng. only (pp)	
	HR	Eng.	Op. Res.	Op. Code
Test Cases (Z)	—	+0.1 (1.5)	+0.7 (1.2)	-1.9 (1.5)
GPA (Z)	-0.4 (1.2)	+0.9 (1.4)	+0.5 (1.4)	-1.6 (1.4)
# Awards (Z)	-0.5 (1.3)	+2.3 (1.8)	+0.2 (1.2)	+0.5 (1.2)
Top Exp.	-3.8 (2.7)	+0.9 (3.5)	-2.0 (2.4)	-0.9 (2.2)
Quality Mismatch I	—	+2.6 (4.6)	+0.4 (3.1)	+2.0 (3.4)
Quality Mismatch II	—	-3.1 (3.3)	+2.1 (2.4)	-3.1 (2.9)

(1) * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; standard errors clustered by respondent.

(2) Left table: baseline treatment effects, one regression per column — time in seconds (display positions 4–18, first three dropped as warm-up), opening in percentage points (all positions).

(3) Right table: heterogeneity — each cell is a separate focused regression that adds one Blinding-by-characteristic interaction to the baseline; none is significant (all $p > 0.22$).

(4) HR cells for test cases and Quality Mismatch are blank (HR never saw the code or tests). Quality Mismatch I/II overlap with GPA, awards, and test by construction.

2.3 Signal Substitution – Re-weighting of Visible Resume Cues

Question: What visible resume signals does the evaluator lean on more (or less) under blinding?

Specification:

$$\text{Overall_Evaluation_Score (Z)}_{ij} \sim \sum_{s \in \mathcal{S}_{\text{role}}} \beta_s \text{Signal}_s^i \times \mathbb{1}[\text{Blind}]_j \mid \text{Resume_ID FE}_i + \text{Respondent_ID FE}_j + \text{Display_Order FE}_{ij}$$

Two-way clustered SEs on (responseid, resume_id).

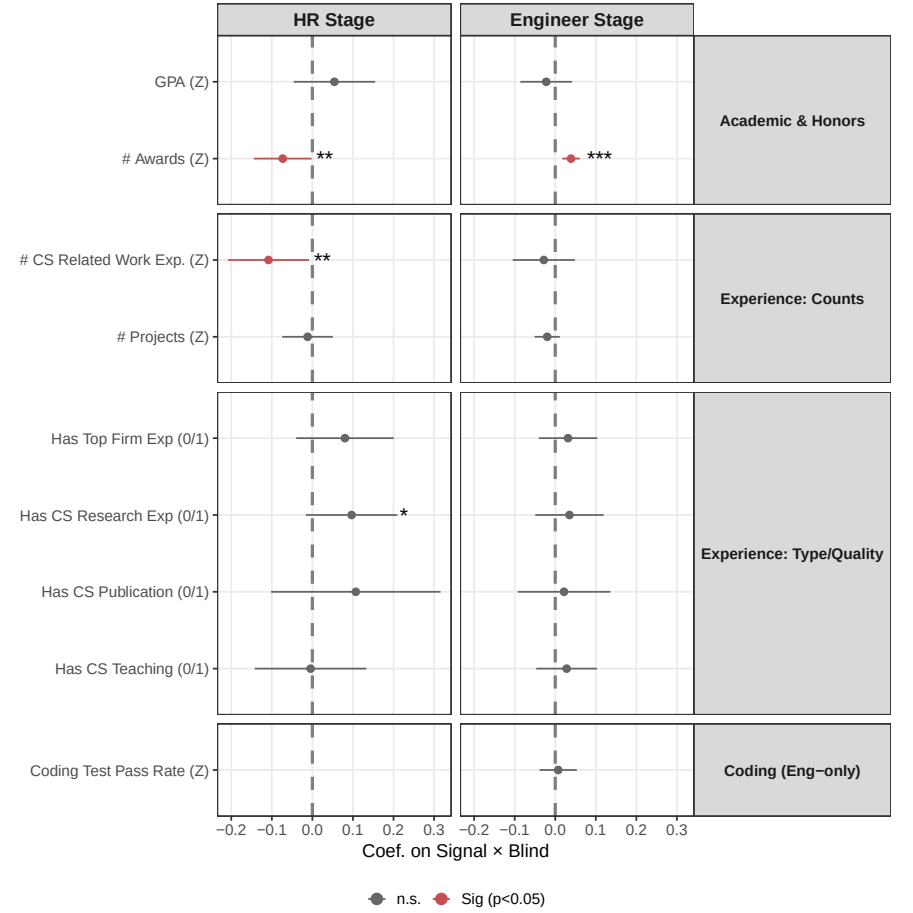
Table 7: Signal Substitution

Blinding: Change in Signal Weights on Evaluator-Assigned Overall Scores

	HR	Engineer
Academic & Honors		
GPA (Z)	0.054 (0.051)	−0.022 (0.032)
# Awards (Z)	−0.073** (0.036)	0.039*** (0.011)
Experience: Counts		
# CS Related Work Exp. (Z)	−0.108** (0.051)	−0.028 (0.039)
# Projects (Z)	−0.012 (0.032)	−0.020 (0.016)
Experience: Type/Quality		
Has Top Firm Exp (0/1)	0.080 (0.061)	0.032 (0.037)
Has CS Research Exp (0/1)	0.097* (0.057)	0.035 (0.043)
Has CS Publication (0/1)	0.107 (0.107)	0.022 (0.058)
Has CS Teaching (0/1)	−0.004 (0.070)	0.028 (0.038)
Coding (Eng-only)		
Coding Test Pass Rate (Z)		0.007 (0.023)
Display Position FE	✓	✓
Resume ID FE	✓	✓
Resume - Coding Test Version FE	×	×
Respondent ID FE	✓	✓
Respondent Age (Cont) and Demographic FEs	×	×
Obs.	2.466	2.754
R^2	0.423	0.749
Adj. R^2	0.378	0.730

Figure 1: Signal Substitution

Blinding: Change in Signal Weights on Evaluator-Assigned Overall Scores



Notes: Coding Test Pass Rate is the number of test cases each applicant's code passed, which was shown to the Engineer evaluators. CS Publication appears on only 2 of the 18 shown resumes, so its weight-shift here (and in Figure 1) is imprecise; its validity in Section 3 uses all 69 students.

2.4 Re-weighting of Internal Rubric – Coding / Fit / Stay Dimension Aggregation

Question: Do the weights evaluators place on the three sub-dimension ratings (Coding Skill, Fit, Stay) shift under blinding?

Specification:

$$\text{Overall_Evaluation_Score (Z)}_{ij} \sim \sum_{d \in \{\text{Coding_Skill, Fit, Stay}\}} (\beta_d + \gamma_d \mathbb{1}[\text{Blind}]_j) \text{Score}_d^i$$

$$\left| \text{Resume_ID FE}_i + \text{Respondent_ID FE}_j + \text{Display_Order FE}_{ij} \right|$$

Two-way clustered SEs on (responseid, resume_id).

Table 8: Internal Rubric Shift

	HR	Engineer
Sub-dimension level (nonblind baseline)		
Coding (Z)	0.490*** (0.046)	0.543*** (0.039)
Fit (Z)	0.267*** (0.047)	0.223*** (0.039)
Stay (Z)	0.139*** (0.031)	0.077*** (0.018)
Re-weighting under Blind Evaluation		
Coding × Blind	0.027 (0.059)	−0.040 (0.056)
Fit × Blind	0.007 (0.055)	0.033 (0.057)
Stay × Blind	−0.064 (0.039)	0.009 (0.027)
Display Position FE	✓	✓
Resume ID FE	✓	✓
Resume - Coding Test Version FE	×	×
Respondent ID FE	✓	✓
Respondent Age (Cont) and Demographic FEs	×	×
Obs.	2.376	2.718
R^2	0.788	0.895
Adj. R^2	0.771	0.887

3 Signal Validity: Are Visible Signals Predictive of True Coding Ability?

Why this matters. §2.2 shows that under blinding, evaluators reweight their attention onto visible resume signals — but a positive weight-shift only counts as rational signal substitution if those signals *actually predict* true productivity. If the signals blinding pushes evaluators toward have zero validity, the §2.2 effect is just *credentialism* (the flight-to-credentials hypothesis). This section provides the validity benchmark.

Question: Among the visible resume signals evaluators rely on, which actually predict true productivity?

Specification:

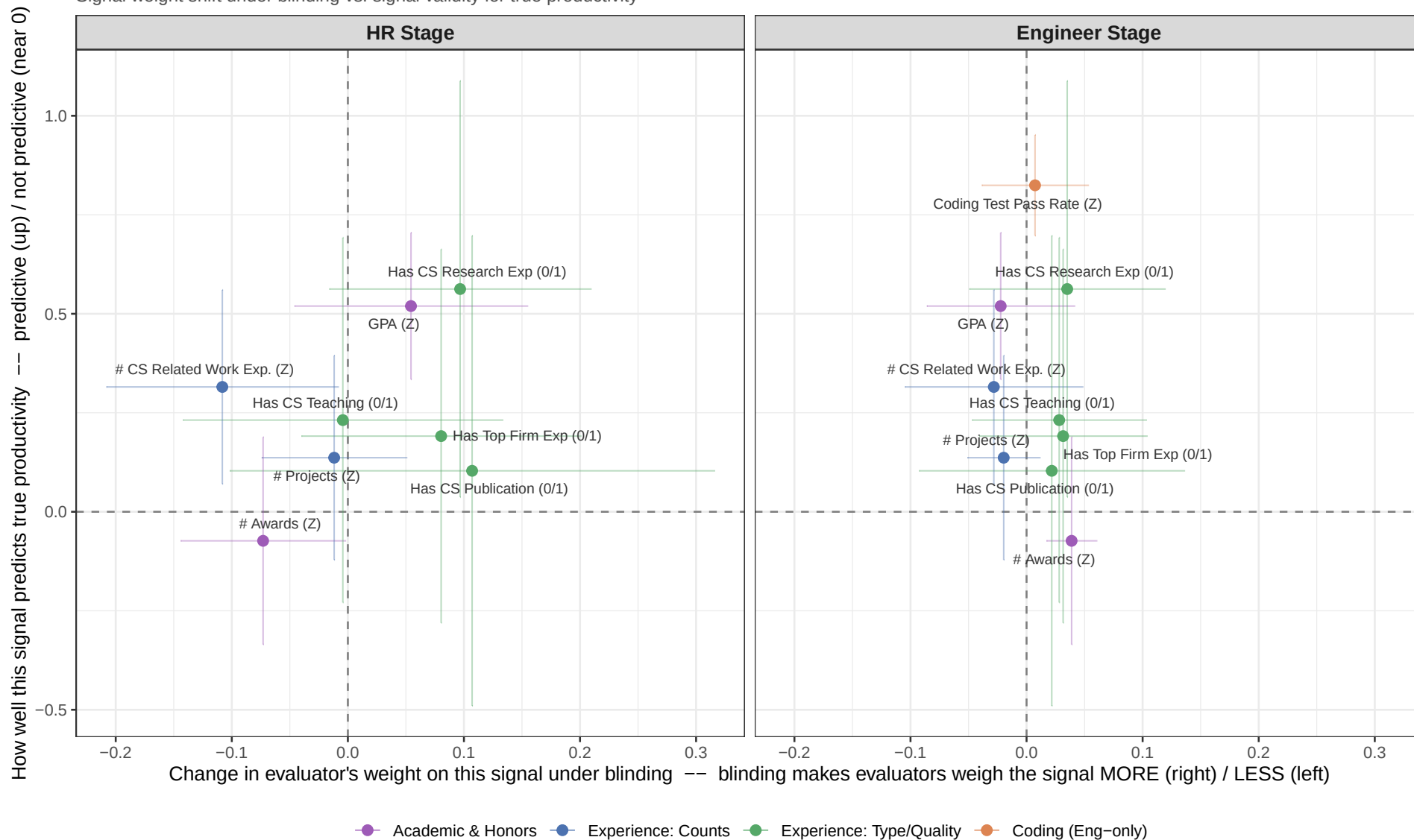
$$\text{True_Productivity_Score (Z)}_i \sim \alpha + \gamma_s \text{Signal}_s^i + \varepsilon_i, \quad i = 1, \dots, 69$$

One univariate OLS per signal over all 69 students; HC1 robust SE.

Table 9: Signal Validity (all 69 students) — univariate slope of true productivity (Z) on each visible signal. Test Cases validated against leave-it-out productivity (readability + efficiency + runtime) to avoid tautology.

Signal	Estimate	R^2	Obs.
Academic & Honors			
GPA (Z)	0.519*** (0.095)	0.268	68
# Awards (Z)	-0.073 (0.134)	0.005	69
Experience: Counts			
# CS Related Work Exp. (Z)	0.315** (0.125)	0.099	69
# Projects (Z)	0.137 (0.132)	0.019	69
Experience: Type/Quality			
Has Top Firm Exp (0/1)	0.191 (0.241)	0.009	69
Has CS Research Exp (0/1)	0.562** (0.268)	0.075	69
Has CS Publication (0/1)	0.103 (0.303)	0.001	69
Has CS Teaching (0/1)	0.232 (0.235)	0.013	69
Coding (Eng-only)			
Coding Test Pass Rate (Z)	0.824*** (0.065)	0.680	69

Signal weight shift under blinding vs. signal validity for true productivity



Top-right quadrant: rational re-weighting (blinding shifts evaluators toward genuinely predictive signals). Top-left: missed opportunity (signal is predictive but evaluators weigh it less under blinding). Bottom-right: fl

4 Individual Response to Institutional Policy (Institutional Crowding Out or Compensatory Bias)

Takeaway / Summary.

- **The Puzzle of Individual Ideology:** The two-way relationship between an evaluator’s personal DEI priority and their evaluation scores is surprisingly *muted and statistically noisy*. At first glance, personal ideology appears to have little impact on how blind hiring reshuffles outcomes.
 - **Compensatory Bias** (Firms with NO Diversity Mandate, see [red lines](#)): When a firm lacks an explicit objective policy targeting URMs, highly pro-DEI HR professionals engage in *compensatory bias* — filling the perceived institutional void by informally inflating minority candidates’ scores in the baseline.
 - **Institutional Crowding-Out** (Firms WITH a Diversity Mandate, see [blue lines](#)): Conversely, when the firm explicitly mandates URM targeting, this formal policy *crowds out* the evaluator’s informal efforts; pro-DEI HR professionals *mute* their premium to under-represented minorities’ scores.
- (Note the *divergence between HR and Engineer results*, particularly in the triple-interaction columns.)

Note on Key Index Creation.

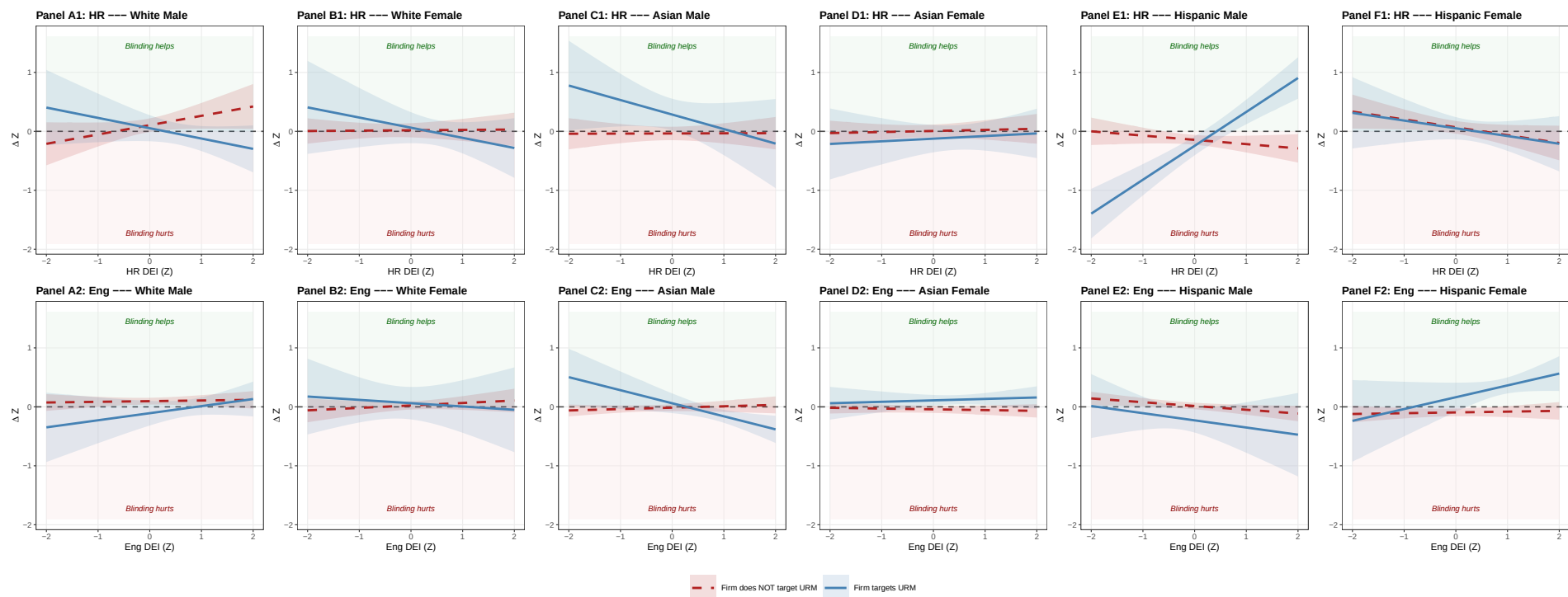
- *Individual DEI Index (Z)*: Evaluator’s personal DEI orientation, condensed into a single z-scored score. (Low = Anti-DEI, High = Pro-DEI)
 - Mathematical expression: $\text{Index} = z\left(\frac{1}{5} \sum_{j=1}^5 z(x_j)\right)$, z-scored within (**role**, **treat**) cells.
 - **pro_dei_race** — stated agreement that race DEI is important in hiring (Likert).
 - **pro_dei_gender** — stated agreement that gender DEI is important in hiring (Likert).
 - **pro_diverse** — stated importance of working at a company with a diverse workforce (Likert).
 - **pro_hiring_dei** — stated belief that DEI initiatives help firms hire better candidates (Likert).
 - **div_priority** — revealed-priority rank of “Diversity and Representation” in the seven-goals question.
- *Firm Targets Fem/His (0/1)*: Indicator for firm policy — whether the evaluator’s firm formally targets Female or Hispanic candidates in hiring.
 - Mathematical expression: $\text{Idx} = \mathbb{I}\{\text{firm_targets_female} = 1 \vee \text{firm_targets_hispanic} = 1\}$.
 - **firm_targets_female** — 1 if option 6 (“Female”) was selected on **s5_policy_diversity** (multi-select).
 - **firm_targets_hispanic** — 1 if option 3 (“Hispanic or Latino”) was selected on the same multi-select.
- *Firm DEI Attn Index (Z)*: Respondent’s perception of how much attention their firm currently pays to race and gender DEI in hiring, condensed into a single z-scored score.
 - Mathematical expression: $\text{Index} = z\left(\frac{1}{2}(z(\text{race_attn}) + z(\text{gender_attn}))\right)$, z-scored within (**role**, **treat**) cells.
 - **s5_view_dei_race_f** — Likert: respondent’s perception of firm’s current attention to race DEI in hiring.
 - **s5_view_dei_gender_f** — Likert: respondent’s perception of firm’s current attention to gender DEI in hiring.

4.1 By Six Demographics

Table 10: DEI Heterogeneity in the Effect of Blinding on Overall Evaluation Score, by Race-Gender (6 cells): Simple, Triple-with-Firm-Targets-URM, Triple-with-Firm-Pays-DEI-Attention.

	Simple DEI Heterogeneity		Triple: Firm Targets URM		Triple: Firm Pays DEI Attention	
	HR	Engineer	HR	Engineer	HR	Engineer
White Male	0.059 (0.053)	0.086*** (0.028)	0.105* (0.059)	0.096*** (0.030)	0.128** (0.062)	0.055* (0.030)
White Female	0.011 (0.052)	0.023 (0.036)	0.018 (0.061)	0.024 (0.036)	-0.001 (0.067)	0.034 (0.036)
Asian Male	0.015 (0.060)	-0.017 (0.028)	-0.036 (0.058)	-0.014 (0.030)	0.072 (0.068)	-0.027 (0.034)
Asian Female	-0.021 (0.048)	-0.030 (0.029)	0.006 (0.055)	-0.042 (0.031)	-0.027 (0.060)	-0.021 (0.041)
Hispanic Male	-0.109** (0.046)	-0.004 (0.029)	-0.144*** (0.043)	0.015 (0.029)	-0.179*** (0.053)	0.028 (0.028)
Hispanic Female	0.064 (0.047)	-0.070** (0.033)	0.068 (0.055)	-0.094*** (0.033)	0.032 (0.055)	-0.081* (0.046)
White Male $\times$ Individual DEI Index (Z)	0.098 (0.079)	0.010 (0.033)	0.158* (0.091)	0.011 (0.034)	0.166** (0.066)	0.040 (0.045)
White Female $\times$ Individual DEI Index (Z)	-0.017 (0.051)	0.037 (0.047)	0.006 (0.056)	0.042 (0.048)	0.000 (0.069)	-0.012 (0.046)
Asian Male $\times$ Individual DEI Index (Z)	-0.005 (0.064)	0.013 (0.026)	0.002 (0.062)	0.024 (0.028)	-0.021 (0.066)	0.043 (0.048)
Asian Female $\times$ Individual DEI Index (Z)	0.009 (0.046)	-0.006 (0.022)	0.018 (0.052)	-0.013 (0.022)	-0.006 (0.059)	0.009 (0.049)
Hispanic Male $\times$ Individual DEI Index (Z)	0.020 (0.062)	-0.075*** (0.029)	-0.072 (0.057)	-0.064** (0.028)	-0.033 (0.069)	-0.095** (0.040)
Hispanic Female $\times$ Individual DEI Index (Z)	-0.135** (0.062)	0.037 (0.034)	-0.133* (0.069)	0.013 (0.033)	-0.152*** (0.056)	0.011 (0.065)
White Male $\times$ Firm Targets Fem/His (0/1)			-0.052 (0.128)	-0.204* (0.110)		
White Female $\times$ Firm Targets Fem/His (0/1)			0.044 (0.149)	0.038 (0.145)		
Asian Male $\times$ Firm Targets Fem/His (0/1)			0.321** (0.148)	0.075 (0.090)		
Asian Female $\times$ Firm Targets Fem/His (0/1)			-0.131 (0.131)	0.152*** (0.055)		
Hispanic Male $\times$ Firm Targets Fem/His (0/1)			-0.101 (0.093)	-0.244** (0.109)		
Hispanic Female $\times$ Firm Targets Fem/His (0/1)			-0.017 (0.112)	0.258** (0.130)		
White Male $\times$ Firm DEI Attn Index (Z)					-0.036 (0.108)	0.011 (0.027)
White Female $\times$ Firm DEI Attn Index (Z)					-0.111 (0.068)	0.033 (0.043)
Asian Male $\times$ Firm DEI Attn Index (Z)					0.129* (0.074)	-0.045 (0.041)
Asian Female $\times$ Firm DEI Attn Index (Z)					-0.017 (0.076)	-0.042 (0.036)
Hispanic Male $\times$ Firm DEI Attn Index (Z)					0.166** (0.069)	0.020 (0.026)
Hispanic Female $\times$ Firm DEI Attn Index (Z)					-0.116 (0.080)	0.032 (0.052)
White Male $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			-0.333** (0.152)	0.109 (0.111)		
White Female $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			-0.178 (0.164)	-0.098 (0.166)		
Asian Male $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			-0.249 (0.192)	-0.245*** (0.090)		
Asian Female $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			0.026 (0.130)	0.037 (0.060)		
Hispanic Male $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			0.648*** (0.106)	-0.058 (0.154)		
Hispanic Female $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			0.002 (0.147)	0.187 (0.124)		
White Male $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					-0.222* (0.125)	0.024 (0.030)
White Female $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					0.014 (0.090)	0.004 (0.041)
Asian Male $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					-0.219** (0.089)	0.002 (0.027)
Asian Female $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					0.037 (0.078)	-0.022 (0.028)
Hispanic Male $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					0.159* (0.085)	-0.045** (0.022)
Hispanic Female $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					0.219** (0.104)	0.041 (0.046)
Display Position FE	✓	✓	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×	×	×
Respondent ID FE	×	×	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓	✓	✓
Observations	2466	2754	2466	2754	2160	2340

Standard errors clustered by respondent.



Marginal-effects “scissors” plot built from the §2.1 columns 3-4 (HR & Engineer triple with Firm Targets Fem/His), per race-gender. Top row: HR evaluators (Panels A1–F1). Bottom row: Engineer evaluators (Panels A2–F2). Lines: predicted effect of blinding on each group’s overall score as a function of the rater’s Individual DEI Index (Z); shaded bands are 95% CIs (SE clustered by respondent).

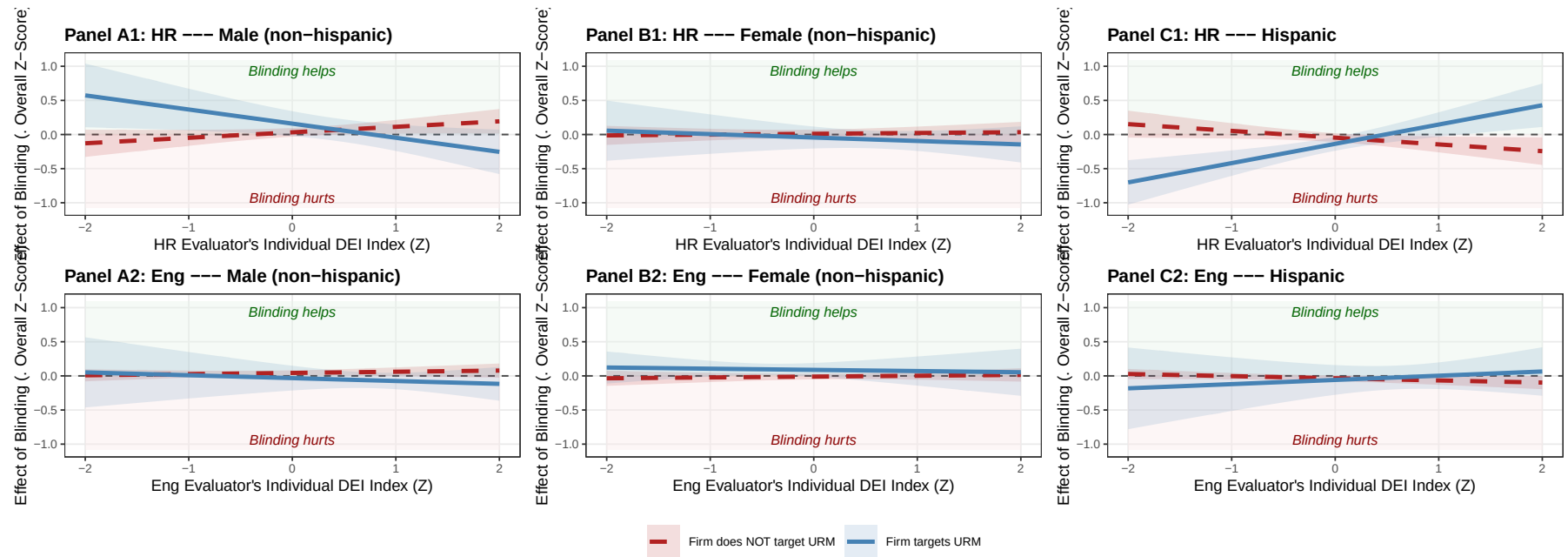
4.2 By Trichotomy (Majority vs Gender Minority vs Racial Minority)

Table 11: DEI Heterogeneity in the Effect of Blinding on Overall Evaluation Score, by URM Axis (Male NH / Female NH / Hispanic): Simple, Triple-with-Firm-Targets-URM, Triple-with-Firm-Pays-DEI-Attention.

	Simple DEI Heterogeneity		Triple: Firm Targets URM		Triple: Firm Pays DEI Attention	
	HR	Engineer	HR	Engineer	HR	Engineer
Male (non-hispanic)	0.036 (0.034)	0.036* (0.019)	0.032 (0.032)	0.043** (0.019)	0.099** (0.038)	0.015 (0.019)
Female (non-hispanic)	-0.007 (0.028)	-0.005 (0.020)	0.011 (0.031)	-0.011 (0.021)	-0.016 (0.034)	0.005 (0.025)
Hispanic	-0.030 (0.032)	-0.033 (0.021)	-0.046 (0.033)	-0.035 (0.021)	-0.084*** (0.032)	-0.022 (0.026)
Male (NH) $\times$ Individual DEI Index (Z)	0.048 (0.043)	0.012 (0.022)	0.080* (0.046)	0.017 (0.023)	0.074** (0.036)	0.042 (0.032)
Female (NH) $\times$ Individual DEI Index (Z)	-0.002 (0.031)	0.014 (0.024)	0.012 (0.034)	0.012 (0.025)	-0.001 (0.037)	0.000 (0.032)
Hispanic $\times$ Individual DEI Index (Z)	-0.048 (0.046)	-0.027 (0.019)	-0.099** (0.048)	-0.031 (0.019)	-0.079* (0.043)	-0.045 (0.038)
Male (NH) $\times$ Firm Targets Fem/His (0/1)			0.127 (0.099)	-0.076 (0.094)		
Female (NH) $\times$ Firm Targets Fem/His (0/1)			-0.055 (0.089)	0.099* (0.055)		
Hispanic $\times$ Firm Targets Fem/His (0/1)			-0.091 (0.061)	-0.024 (0.113)		
Male (NH) $\times$ Firm DEI Attn Index (Z)					0.045 (0.055)	-0.017 (0.023)
Female (NH) $\times$ Firm DEI Attn Index (Z)					-0.064 (0.039)	-0.005 (0.025)
Hispanic $\times$ Firm DEI Attn Index (Z)					0.020 (0.052)	0.025 (0.026)
Male (NH) $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			-0.287*** (0.102)	-0.059 (0.094)		
Female (NH) $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			-0.062 (0.089)	-0.030 (0.075)		
Hispanic $\times$ Individual DEI Index (Z) $\times$ Firm Targets Fem/His (0/1)			0.382*** (0.093)	0.093 (0.114)		
Male (NH) $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					-0.221*** (0.069)	0.013 (0.022)
Female (NH) $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					0.027 (0.055)	-0.009 (0.022)
Hispanic $\times$ Individual DEI Index (Z) $\times$ Firm DEI Attn Index (Z)					0.195*** (0.071)	-0.004 (0.023)
Display Position FE	✓	✓	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×	×	×
Respondent ID FE	×	×	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓	✓	✓
Observations	2466	2754	2466	2754	2160	2340

* p < 0.1, ** p < 0.05, *** p < 0.01

Standard errors clustered by respondent.



Marginal-effects “scissors” plot built from the §2.2 columns 3-4 (HR & Engineer triple with Firm Targets Fem/His). Top row: HR evaluators (Panels A1, B1, C1). Bottom row: Engineer evaluators (Panels A2, B2, C2). Lines: predicted effect of blinding on each group’s overall score as a function of the rater’s Individual DEI Index (Z); shaded bands are 95% CIs from the linear combination of axis-triple coefficients (SE clustered by respondent).

5 Inter-Departmental Backlash and Compensation: Signed Directional Index

Spec. Triple-interaction model with a **signed, DEI-specific, directional** perception index per role: positive end captures the role-appropriate “misalignment in the bad direction” (HR overreach for Eng; Eng undercare for HR); negative end captures the opposite direction. Both directions are continuous, not thresholded.

Survey directional structure (used directly). The multi-select options `aligndiff{role}_1..6` record which DEI dimensions the respondent flagged, and the items themselves are directional:

- Options 1 / 4: counterpart places *higher* priority on gender / racial diversity (rebellion-relevant for Eng; over-care-relevant for HR).
- Options 2 / 5: counterpart places *lower* priority on gender / racial diversity (under-care-relevant for HR; opposite-direction signal for Eng).
- Options 3 / 6: counterpart places priority on *different* groups (no direction — ignored here).

Index construction (per role; respondent-level then merged back):

- **Eng (HR Overreach direction).** Likert: `s5_hr_influence` (1–4, “to what extent does HR push DEI in hiring”; higher = more push). Directional count: $(\text{aligndiffeng}_1 + \text{aligndiffeng}_4) - (\text{aligndiffeng}_2 + \text{aligndiffeng}_5)$ (positive = HR cares more on race/gender DEI; negative = HR cares less). Composite: `eng_dir = s5_hr_influence + 0.5 · directional_count`, then z-scored within Eng.
- **HR (Eng Undercare direction).** Likert: `-s5_engineer_care` (negated 0–4 “to what extent does Eng prioritize DEI”; higher negated = Eng cares less). Directional count: $(\text{aligndiffhr}_2 + \text{aligndiffhr}_5) - (\text{aligndiffhr}_1 + \text{aligndiffhr}_4)$ (positive = Eng cares less on race/gender DEI; negative = Eng cares more). Composite: `hr_dir = -s5_engineer_care + 0.5 · directional_count`, then z-scored within HR.
- `misalign_z = eng_dir_z` for Eng rows / `hr_dir_z` for HR rows, used as `perc_role` in the regressions. Sample inclusion: drop only respondents NA on the Likert anchor (`s5_hr_influence` for Eng / `s5_engineer_care` for HR).

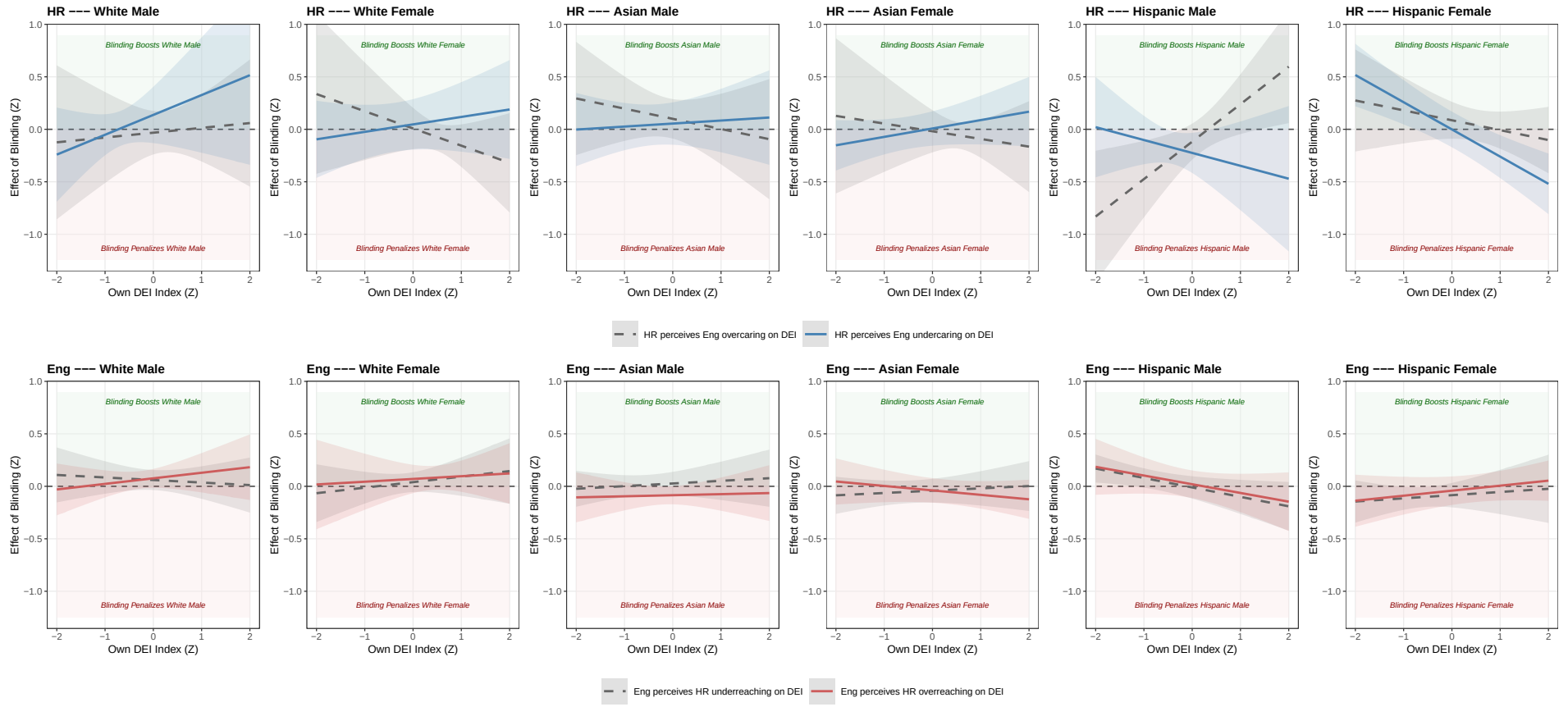
Substantive interpretation. `perc_role > 0` = rebellion conditions'' for Eng /compensation conditions'' for HR; `perc_role < 0` = the opposite direction (Eng who think HR underreaches; HR who think Eng overcares). Triple-interaction slopes can therefore be read as: *how does the effect of blinding on group X change as the rater moves from “counterpart cares too little on DEI” to “counterpart cares too much on DEI”?*

5.1 By Six Demographics

	Simple Dir Het.		Triple: Dir x Own DEI	
	HR	Engineer	HR	Engineer
White Male	0.016 (0.071)	0.086*** (0.027)	0.052 (0.095)	0.068** (0.030)
White Female	-0.020 (0.066)	0.050 (0.038)	0.027 (0.076)	0.055 (0.045)
Asian Male	0.051 (0.051)	-0.032 (0.035)	0.077 (0.053)	-0.029 (0.041)
Asian Female	-0.036 (0.051)	-0.055 (0.034)	-0.006 (0.066)	-0.040 (0.036)
Hispanic Male	-0.083 (0.060)	0.005 (0.050)	-0.171** (0.079)	0.005 (0.050)
Hispanic Female	0.081 (0.067)	-0.058 (0.042)	0.042 (0.069)	-0.063 (0.052)
White Male × Dir Idx (Z)	0.008 (0.055)	0.022 (0.049)	0.085 (0.077)	0.008 (0.036)
White Female × Dir Idx (Z)	-0.008 (0.064)	0.030 (0.033)	0.019 (0.082)	0.015 (0.039)
Asian Male × Dir Idx (Z)	-0.032 (0.081)	-0.050 (0.032)	-0.023 (0.084)	-0.056* (0.030)
Asian Female × Dir Idx (Z)	-0.019 (0.053)	-0.009 (0.033)	0.013 (0.066)	0.001 (0.046)
Hispanic Male × Dir Idx (Z)	-0.002 (0.051)	-0.022 (0.034)	-0.054 (0.049)	0.015 (0.035)
Hispanic Female × Dir Idx (Z)	0.060 (0.056)	0.038 (0.039)	-0.044 (0.055)	0.021 (0.038)
White Male × Own DEI Index (Z)			0.118 (0.121)	0.014 (0.061)
White Female × Own DEI Index (Z)			-0.046 (0.109)	0.039 (0.057)
Asian Male × Own DEI Index (Z)			-0.034 (0.082)	0.018 (0.047)
Asian Female × Own DEI Index (Z)			0.003 (0.083)	-0.010 (0.034)
Hispanic Male × Own DEI Index (Z)			0.117 (0.104)	-0.087** (0.039)
Hispanic Female × Own DEI Index (Z)			-0.177*** (0.059)	0.039 (0.041)
White Male × Dir Idx (Z) × Own DEI Index (Z)			0.072 (0.107)	0.039 (0.024)
White Female × Dir Idx (Z) × Own DEI Index (Z)			0.117* (0.061)	-0.013 (0.055)
Asian Male × Dir Idx (Z) × Own DEI Index (Z)			0.063 (0.078)	-0.007 (0.030)
Asian Female × Dir Idx (Z) × Own DEI Index (Z)			0.077 (0.074)	-0.032 (0.027)
Hispanic Male × Dir Idx (Z) × Own DEI Index (Z)			-0.240** (0.098)	0.004 (0.033)
Hispanic Female × Dir Idx (Z) × Own DEI Index (Z)			-0.082 (0.054)	0.009 (0.036)
Display Position FE	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×
Respondent ID FE	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓
Observations	2034	2124	2034	2124
R ²	0.433	0.739	0.441	0.740
Adj. R ²	0.414	0.730	0.418	0.730
R ² Within	0.003	0.007	0.016	0.011
Adj. R ² Within	-0.004	0.000	0.002	-0.002

Dir Idx (Z) = signed directional misalignment index: for Eng, HR Overreach direction (Likert + 0.5 times signed multi-select); for HR, Eng Undercare direction. Z-scored within role. Positive = rebellion conditions for Eng / compensation conditions for HR; negative = opposite. n = 113 HR, 118 Eng. Coefs flipped post-fit so positive = effect of blinding raises the score. Two-way cluster on responseid and resume index.

Dir Idx (signed) $\times$ Own DEI Index — Per Race-Gender (6 cells)



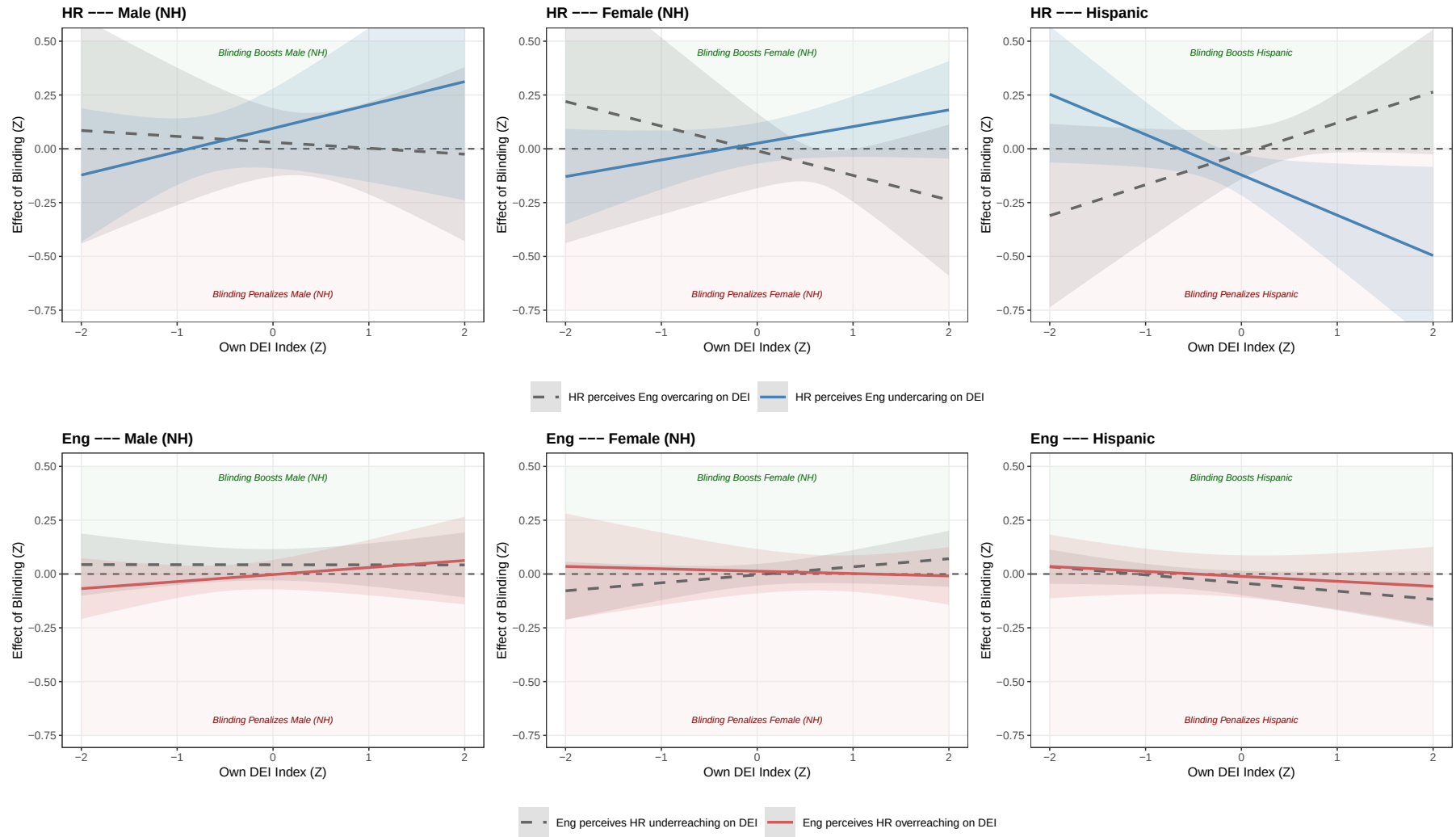
Top row: HR (high Dir = perceives Eng undercare = compensation conditions). Bottom row: Eng (high Dir = perceives HR overreach = rebellion conditions). Bands = 95% CI.

5.2 By Trichotomy (Majority vs Gender Minority vs Racial Minority)

	Simple Dir Het.		Triple: Dir x Own DEI	
	HR	Engineer	HR	Engineer
Male (NH)	0.033 (0.043)	0.028 (0.020)	0.063 (0.059)	0.020 (0.024)
Female (NH)	-0.029 (0.041)	-0.006 (0.025)	0.008 (0.055)	0.005 (0.033)
Hispanic	-0.005 (0.044)	-0.024 (0.030)	-0.072 (0.049)	-0.026 (0.033)
Male (NH) $\times$ Dir Idx (Z)	-0.011 (0.055)	-0.013 (0.034)	0.033 (0.065)	-0.023 (0.027)
Female (NH) $\times$ Dir Idx (Z)	-0.013 (0.036)	0.010 (0.017)	0.017 (0.046)	0.008 (0.025)
Hispanic $\times$ Dir Idx (Z)	0.025 (0.029)	0.004 (0.029)	-0.049** (0.023)	0.015 (0.024)
Male (NH) $\times$ Own DEI Index (Z)			0.041 (0.084)	0.016 (0.035)
Female (NH) $\times$ Own DEI Index (Z)			-0.019 (0.079)	0.013 (0.027)
Hispanic $\times$ Own DEI Index (Z)			-0.022 (0.072)	-0.030 (0.023)
Male (NH) $\times$ Dir Idx (Z) $\times$ Own DEI Index (Z)			0.068 (0.068)	0.017 (0.013)
Female (NH) $\times$ Dir Idx (Z) $\times$ Own DEI Index (Z)			0.096 (0.056)	-0.024 (0.026)
Hispanic $\times$ Dir Idx (Z) $\times$ Own DEI Index (Z)			-0.166*** (0.053)	0.007 (0.018)
Display Position FE	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×
Respondent ID FE	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓
Observations	2034	2124	2034	2124
R ²	0.432	0.737	0.436	0.738
Adj. R ²	0.415	0.730	0.417	0.729
R ² Within	0.001	0.001	0.009	0.002
Adj. R ² Within	-0.003	-0.003	0.000	-0.005

Dir Idx (Z) = signed directional misalignment index. n = 113 HR, 118 Eng. Coefs flipped post-fit so positive = effect of blinding raises the score. Two-way cluster on responseid and resume index.

Dir Idx (signed) $\times$ Own DEI Index — URM-Axis Trichotomy



Top row: HR (high Dir = compensation conditions). Bottom row: Eng (high Dir = rebellion conditions). Bands = 95% CI.

6 Treatment Effect on Productivity of Hired (The Efficiency Myth)

6.1 Does Blinding Lead to More Accurate Coding Skill Evaluations?

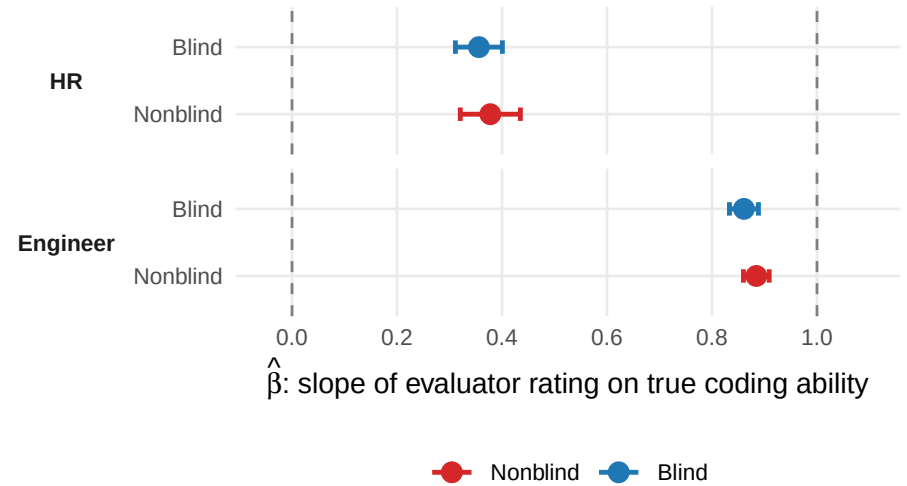
Takeaway: Removing demographic information (or noise) does *not* meaningfully improve evaluators' ability to identify top-ability candidates — signal extraction remains flat.

Specification:

Hiring Evaluation Score (Z) $\sim$ True Coding Ability (Z) $\times$ $\mathbb{1}[\text{Blind arm}]$ | Respondent FE + Display Position FE.

- $H_0: \beta = 0$ — the evaluator extracts *no signal* from true coding ability.
- $H_0: \beta = 1$ — the evaluator is *perfectly calibrated*: a 1 SD rise in true ability produces exactly a 1 SD rise in the rating, with no shrinkage.

	HR	Engineer
$\hat{\beta}_{\text{nonblind}}$ (slope, nonblind arm)	0.378 (0.029)	0.884 (0.012)
$\hat{\beta}_{\text{blind}}$ (slope, blind arm)	0.356 (0.023)	0.861 (0.014)
Observations	2,466	2,754
R^2	0.142	0.763
Adj. R^2	0.085	0.748
Resume (Content) FE	×	×
Display Position FE	✓	✓
Respondent FE	✓	✓
Respondent Characteristics	×	×

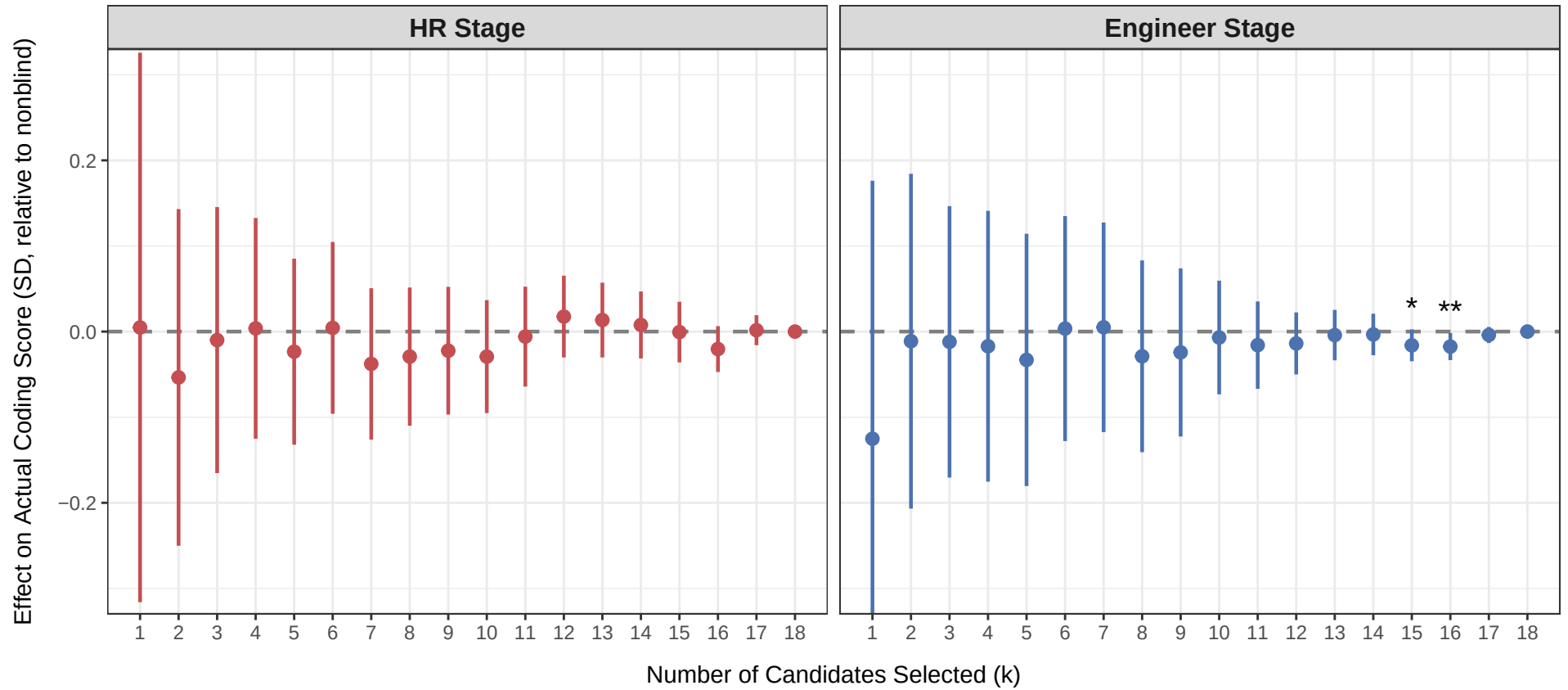


Cluster-robust standard errors (by respondent) shown in parentheses next to each $\hat{\beta}$. Stars on p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Does Blinding Lead to Selection of More Productive Coders?

Takeaway: Blind hiring has a *null* effect on the actual coding quality of the hired cohort.

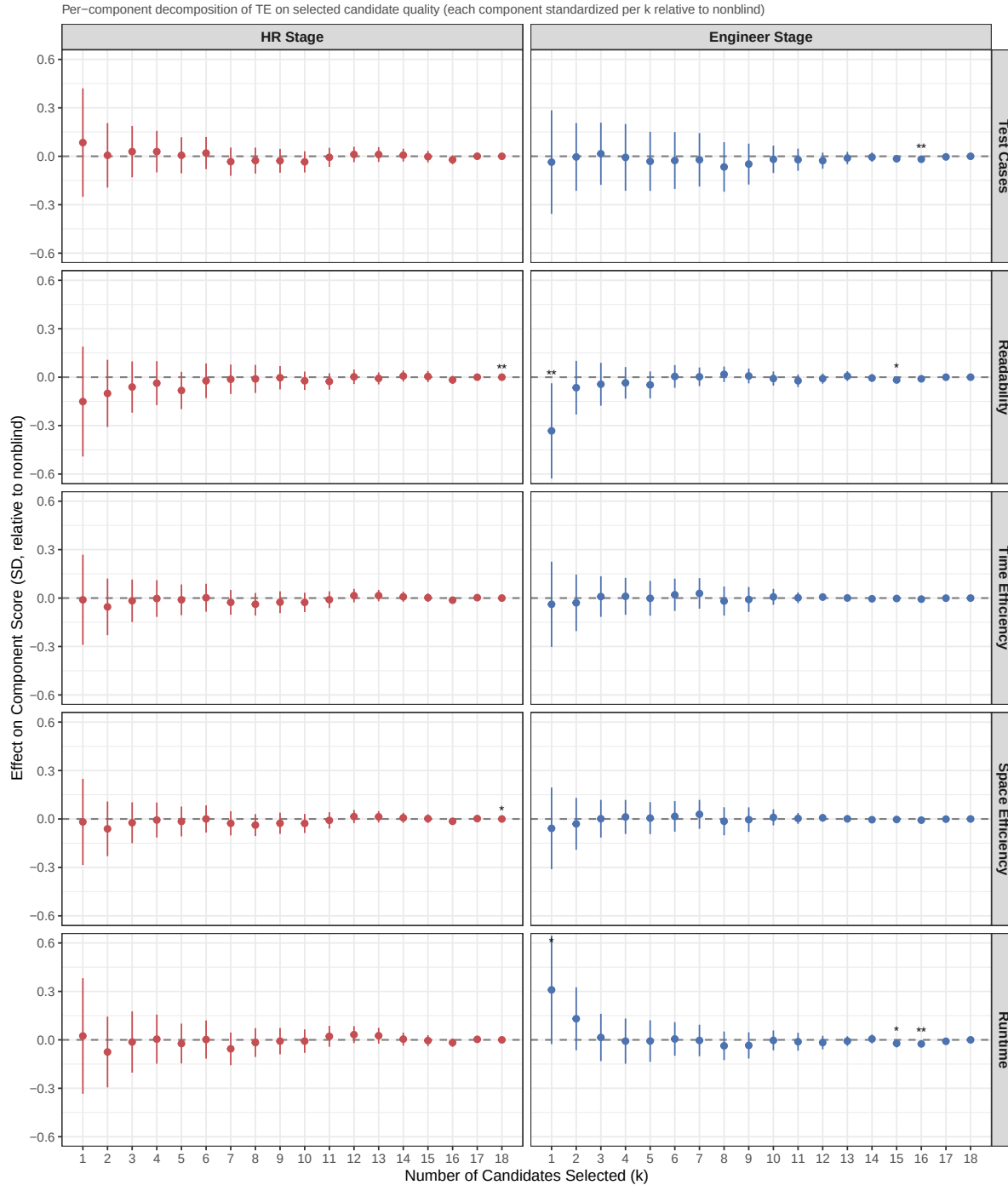
Treatment effect on true productivity measure (coding lab evaluation) of selected candidates, standardized relative to selected candidates under no blind



Positive = blind hiring selects candidates with higher actual coding scores.

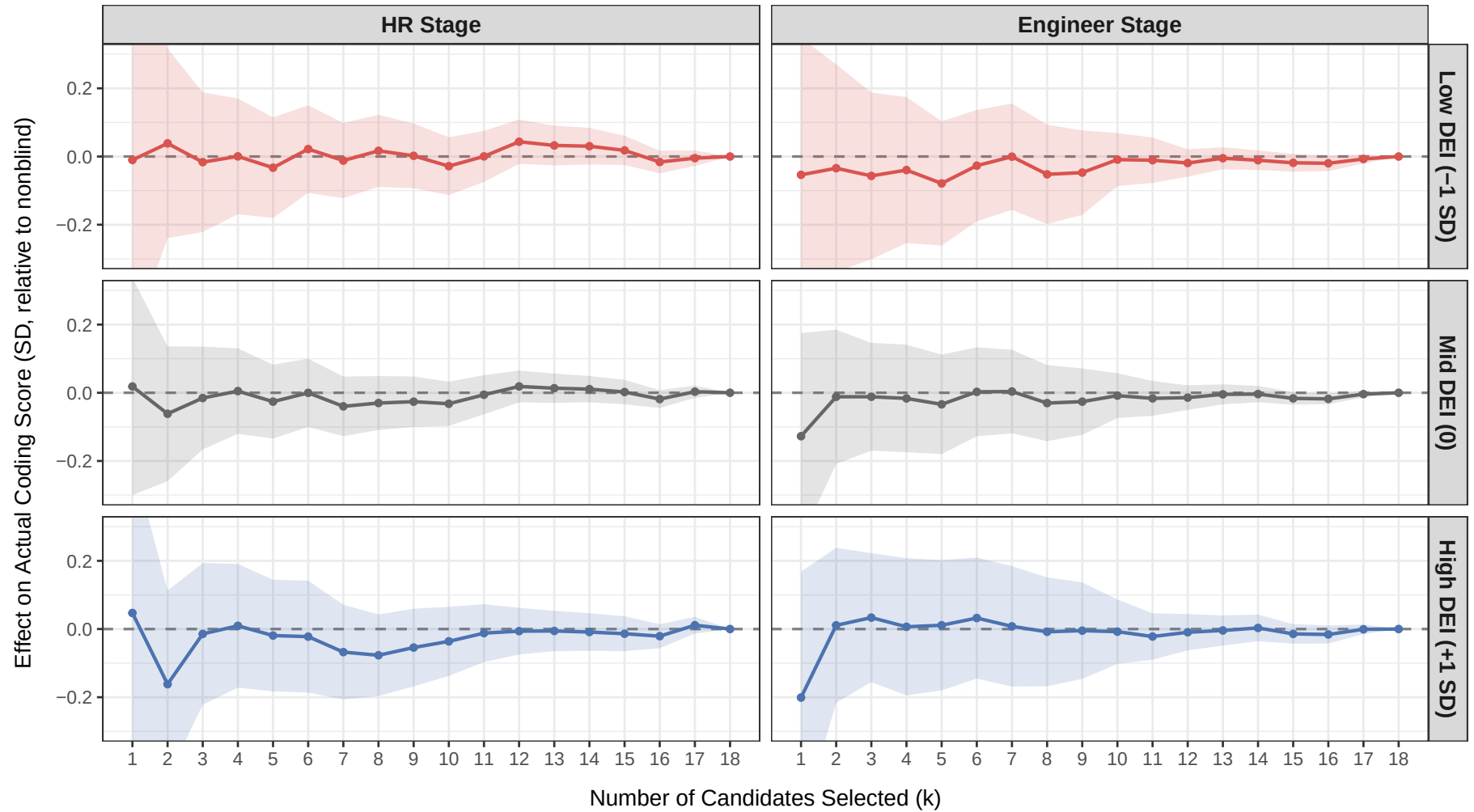
Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

6.2.1 Decomposition of Treatment Effect on Productivity



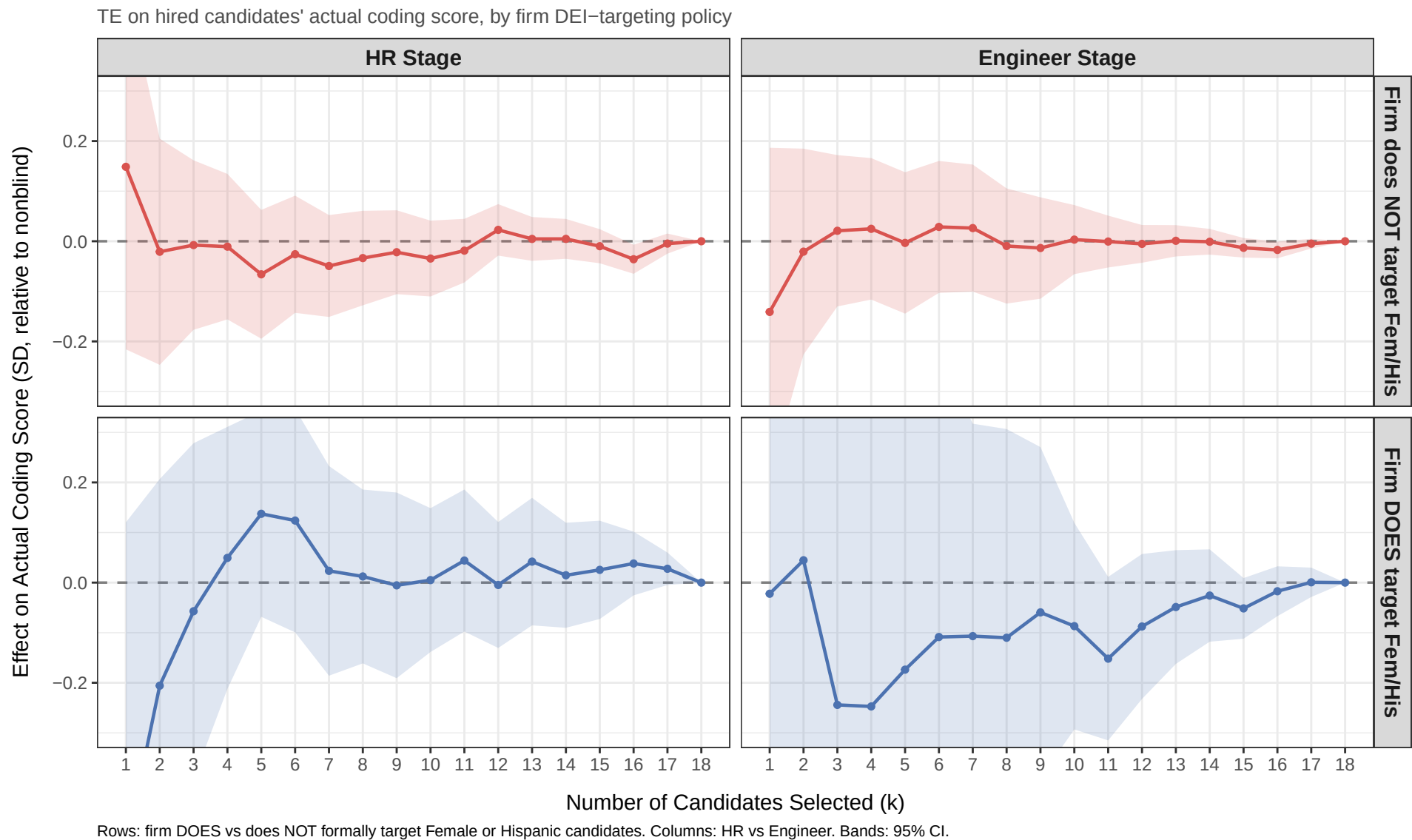
6.3 Does Blinding's Productivity Effect Differ by Evaluator's DEI Orientation?

TE on hired candidates' actual coding score, by respondent's DEI index level

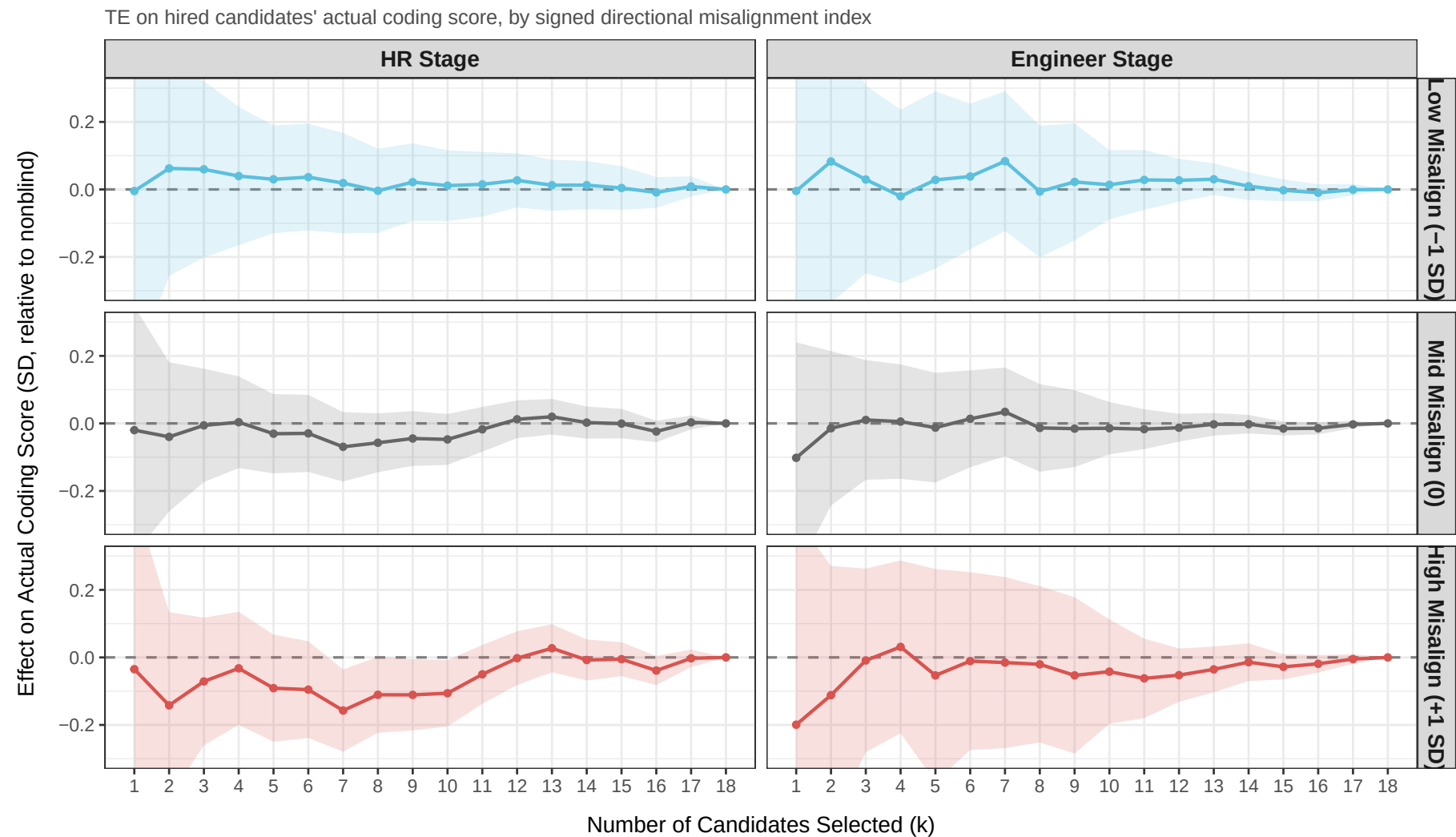


Rows: TE evaluated at -1, 0, +1 SD of the within-role z-scored DEI index. Columns: HR vs Engineer. Bands: 95% CI from the linear combination of treat::blind and treat::blind:dei_index_z.

6.4 Does Blinding’s Productivity Effect Differ by Firm DEI-Targeting Policy?



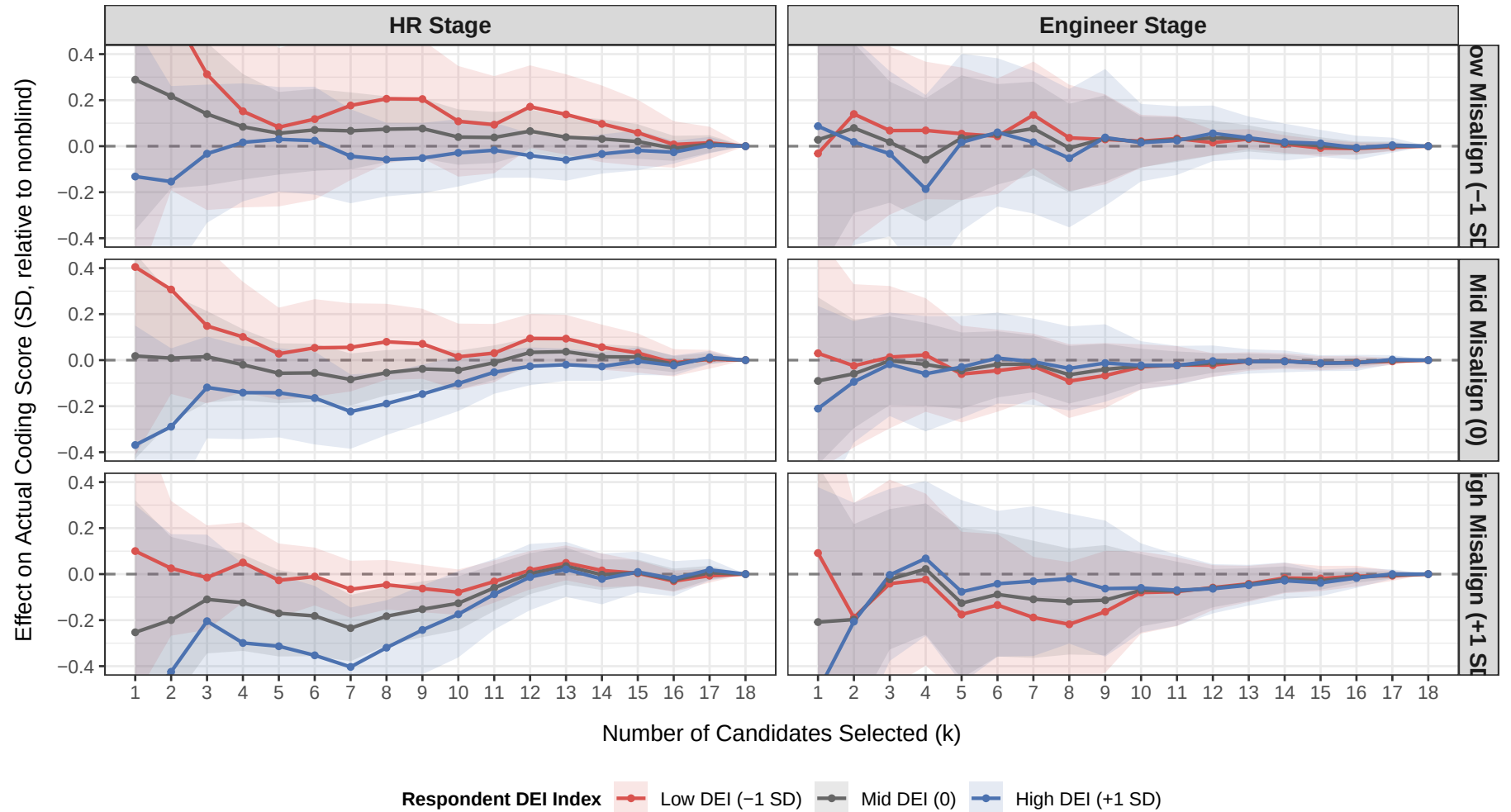
6.5 Does Blinding’s Productivity Effect Differ by Inter-Departmental DEI Misalignment?



Rows: TE evaluated at -1, 0, +1 SD of the within-role z-scored signed directional misalignment index from §3. Columns: HR vs Engineer. Bands: 95% CI.

6.6 Does Blinding's Productivity Effect Vary Jointly by DEI Orientation and Inter-Departmental Misalignment?

TE on hired candidates' actual coding score, by DEI index x signed directional misalignment



Rows: misalignment index at -1/0/+1 SD. Columns: HR vs Engineer. Within panel: respondent DEI index at -1/0/+1 SD. Bands: 95% CI from v' Sigma v with v selecting (treat::blind, treat::blin

7 Out-of-Sample Reweighting to the 69-Student Applicant Pool

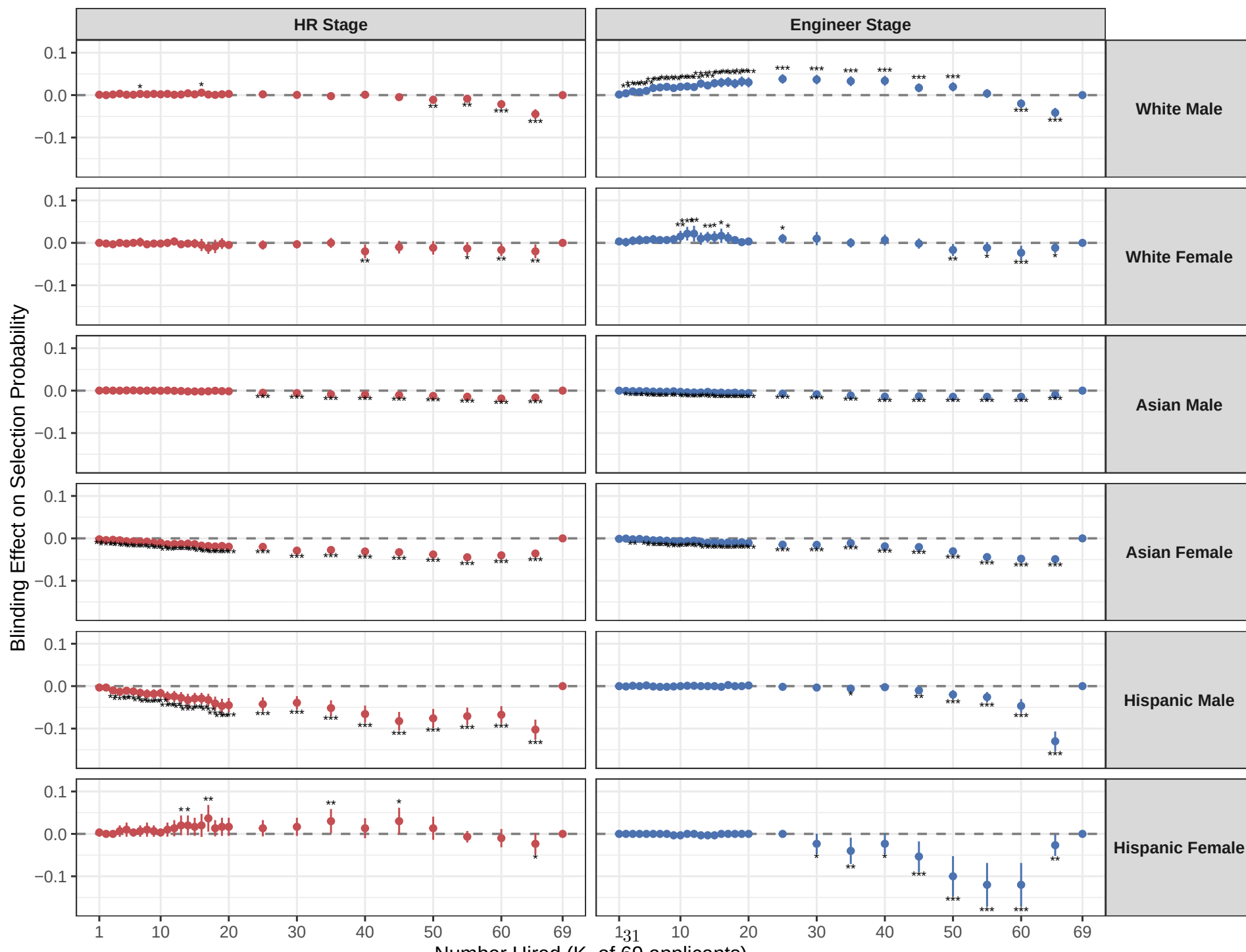
The experiment randomizes demographic labels onto fixed resume content, so assigned demographics are *orthogonal to true productivity by construction* — demographic bias can only reshuffle candidates *within* a productivity level, never systematically select less-productive people. The measured null effect on productivity is therefore largely *mechanical*. This section asks whether that null would survive a realistic applicant pool: we take the measured non-blind demographic biases and apply them to the true 69-student pool (real demographics, real lab productivity), simulating blind vs. non-blind selection at every hiring cutoff $K = 1, \dots, 69$. Behavior is held fixed (the composition channel); this is a *calibration, not an estimate*. The complementary statistical-discrimination channel — the only route by which blinding could *hurt* efficiency — is not identified by this design and is left aside.

7.1 Compositional Change of the Hired Cohort

Blinding removes the non-blind premia/penalties, so it mechanically reshuffles *who* is hired. Effects concentrate deeper in the distribution (higher K), where lower-ranked groups finally enter the hired set, and vanish as $K \rightarrow 69$ (hiring everyone leaves no selection).

Blinding Effect on Selection Probability by Group (Out-of-Sample 69-Pool)

Blinding effect = $P(\text{selected} \mid \text{blind}) - P(\text{selected} \mid \text{non-blind})$. Positive = more likely under blinding.

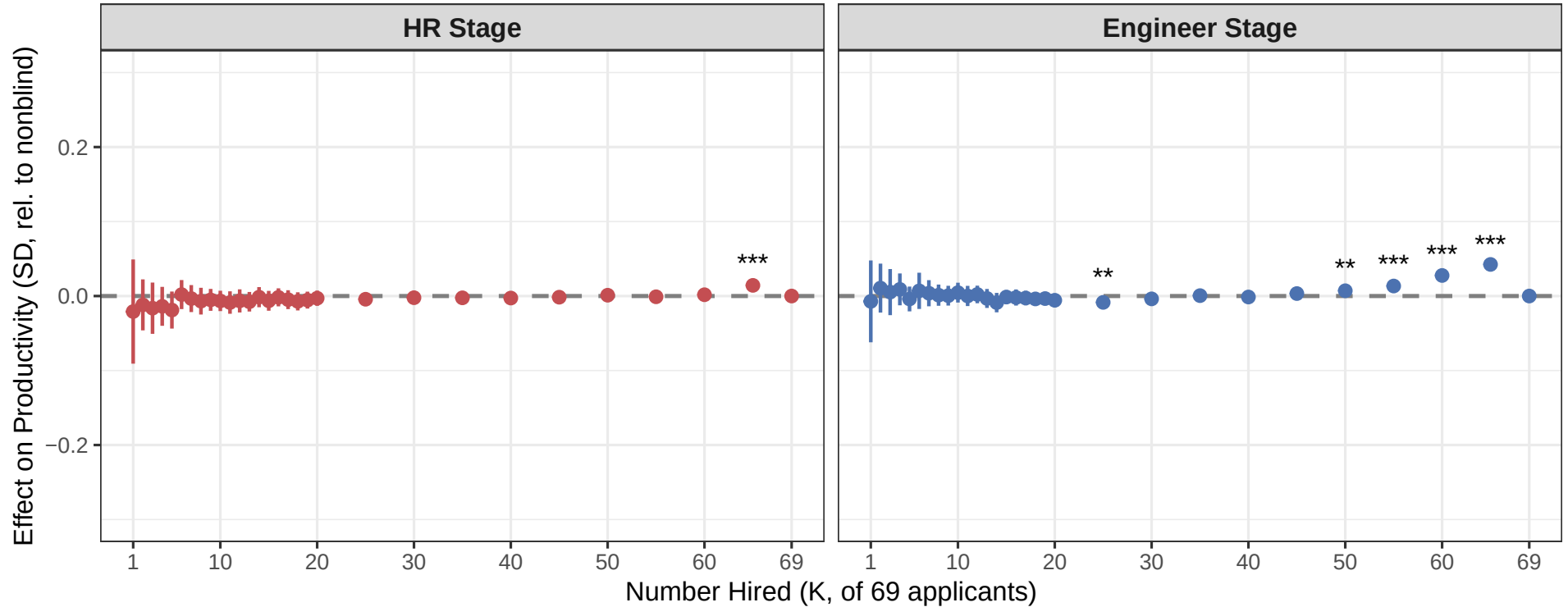


7.2 Treatment Effect on Productivity

Despite the compositional reshuffle, the effect on the true productivity of the hired cohort is **null across the entire hiring margin**, in both stages. Composition moves; productivity does not — the biases are modest and demographic identity carries little productivity information beyond the visible signals.

Treatment Effect on Productivity of Hired (Out-of-Sample 69-Pool)

Effect of blinding on true productivity of the hired cohort, standardized to non-blind at each k



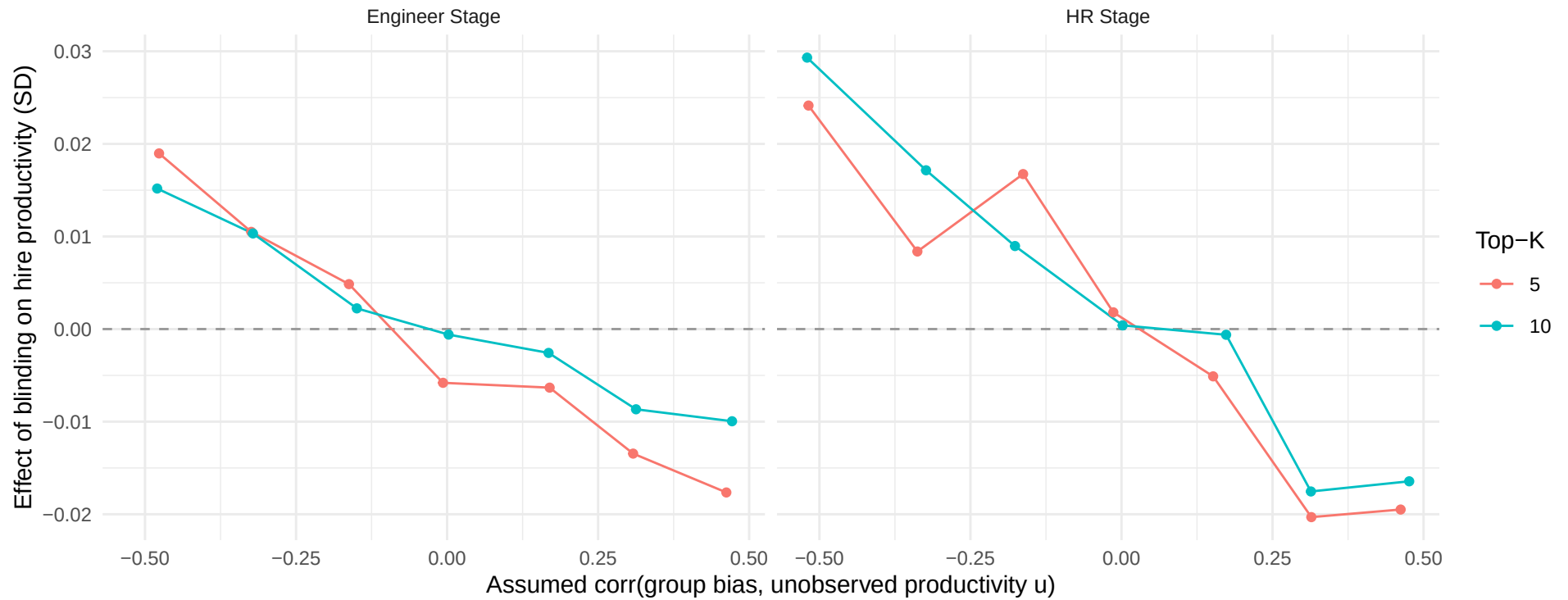
Positive = blinding selects higher-productivity candidates. Fixed-behavior composition channel.

7.3 How Large Would the Identity-Productivity Link Have to Be?

We sweep the correlation between the group bias a candidate attracts and the productivity left *unobserved* by the score. At correlation zero the measured null is recovered; even at a strong correlation the productivity effect stays within roughly ± 0.02 SD. The composition channel is bounded by the (modest) biases and by how little of productivity is unobserved — not by the demographic mix, so reweighting to a different pool cannot rescue a large effect. Any larger effect would have to come from the statistical-discrimination channel, which this design cannot identify.

Sensitivity: productivity effect vs. an assumed identity–productivity correlation

corr = 0 recovers the measured null; even at a strong correlation the effect stays small



Caveats. Calibration, not estimation; behavior held fixed (statistical-discrimination channel omitted); the content-score map is extrapolated from the 18 shown resumes; the pool is Berkeley-CS (79% Asian, with rare Hispanic-female / White-female cells, so those group panels are noisy), not the US market.